



Pranjal Choudhury

Ph.D. in Physics, IIT Guwahati |
Specializing in Optics, Super-Resolution Microscopy, and
Computational Imaging

✉ p.choudhury@iitg.ac.in

✉ pranjal.choudhury264@gmail.com

🐙 GitHub: github.com/pranjal264

🆔 ORCID: [0000-0003-4883-7005](https://orcid.org/0000-0003-4883-7005)

🌐 Website [in](#) [LinkedIn](#)

Education

Degree	Institution	CGPA/%	Year
Ph.D. in Physics	IIT Guwahati	10.00	2026
Integrated M.Sc. in Physics	Tezpur University	8.73 (Rank 1)	2019
Senior Secondary	AHSEC	87.0%	2014
Secondary	CBSE	10.0	2012

Research Experience

Ph.D. Research Student — IIT Guwahati

2020 – 2026

Thesis: Improved Localisation and Reconstruction in a Single Molecule Localisation Microscope to Enable Super-Resolution Imaging with Lowered Resource Pre-requisites.

Advisor: Prof. Bosanta Ranjan Boruah

Work done during PhD:

- **Software Development:** Built custom Python-based SMLM image reconstruction software and integrated it as a Micro-Manager plugin for real-time visualization.
- **Instrumentation:** Collaboratively built a low-cost SMLM setup, including laser alignment and spatial light modulator (SLM) integration.
- **Hardware–Software Interfacing:** Developed an image-based lateral drift correction algorithm for fluorescence microscopes, achieving nanometer-level stability.
- **Deep Learning:** Developed a CNN-based approach for accurate PSF detection in high-density emitter environments. Developed a deep learning-based wavefront reconstruction method for Shack-Hartmann type wavefront sensors.

Visiting Ph.D. Researcher — Imperial College London

May – August 2024

Global Development Hub Fellowship

Supervisor: Prof. Paul M.W. French

- Collaborated on the development of a robust hardware-based autofocus system for focus stabilization in Single Molecule Localization Microscopy (SMLM).

Publications

1. **P. Choudhury**, et al. “Real-Time Lateral Drift Stabilization in Single Molecule Localization Microscopy Using Live Image Re-Registration.” *Journal of Microscopy* (under revision, 2026).
2. **N. Kumar**, et al. “Deep Learning-Driven Wavefront Sensing for Grating-Array-Based Wavefront Sensor.” *IEEE Sensors Journal* (2025). [DOI](#)

3. **P. Choudhury**, et al. "Neural network-assisted localization of clustered point spread functions in single-molecule localization microscopy." *Journal of Microscopy* (2024). [DOI](#)
4. **A. Kumar**, et al. "Tuning the excitation laser power in a stochastic optical reconstruction microscope for Alexa Fluor 647 dye in Vectashield mounting media." *Review of Scientific Instruments* (2024). [DOI](#)
5. **P. Choudhury**, et al. "Adaptive image thresholding and localization of point spread functions with enhanced precision for single molecule localization based super-resolution microscopy." *Optics and Lasers in Engineering* (2024). [DOI](#)
6. **P. Choudhury**, et al. "Localization and Image Reconstruction in a STORM Based Super-resolution Microscope." *Image Processing Online* (2024). [DOI](#)

Teaching & Mentorship

- **Teaching Assistant:** Engineering Optics, Computational Physics, Advanced Physics Laboratory.
- **Mentorship:** Mentored undergraduate and postgraduate students, as well as summer interns, in research projects involving optics, microscopy, and computational imaging.
- Organized scientific talks and outreach programs for SPIE.

Technical Skills

- **Programming:** Python, MATLAB, C, C++, Fortran, Java.
- **Software:** Micro-manager, ImageJ, LabVIEW, Zemax OpticStudio(learning), Mathematica, MS Office.
- **Frameworks:** TensorFlow, PyTorch, Scikit-Learn, Scikit-Image, OpenCV, Napari.
- **Instrumentation:** Custom-built fluorescence optical microscopes, single-molecule localization setups, spatial light modulators, autofocus modules.

Relevant Coursework

M.Sc. Physics: Mathematical Physics, Classical Mechanics, Statistical Mechanics, Quantum Mechanics, Atomic & Molecular Physics, Nuclear & Particle Physics, Condensed Matter Physics.

Ph.D. Coursework: Laser Physics, Nonlinear Optics, Fourier and Guided Wave Optics, Computational Physics, Quantum Optics.

Leadership & Responsibilities

- Assistant Prefect, Patkai Men's Hostel, Tezpur University. (2017 - 2019).
- Member, Research Scholar's Forum, Department of Physics, IIT Guwahati. (2023 - 2025)

Awards & Achievements

- INSPIRE Scholarship for Higher Education, DST. (2014-2019)
- Institute Gold Medal, Integrated M.Sc., Tezpur University. (2019)
- SLET, GATE (PH). (2019)
- CSIR JRF/NET, GATE (PH), JEST and INSPIRE Fellowship. (2020)
- Global Development Hub Fellowship, Imperial College London. (2024)
- Optica Student Leadership (2025).

Professional Memberships

- Student Member, SPIE.
- Student Member, OPTICA.
- Life Member, Physics Association of North East.

References

- Prof. Bosanta Ranjan Boruah
Department of Physics, Indian Institute of Technology Guwahati
✉ brboruah@iitg.ac.in
- Prof. Paul M.W. French
Department of Physics, Imperial College London
✉ paul.french@imperial.ac.uk